INCH-POUND

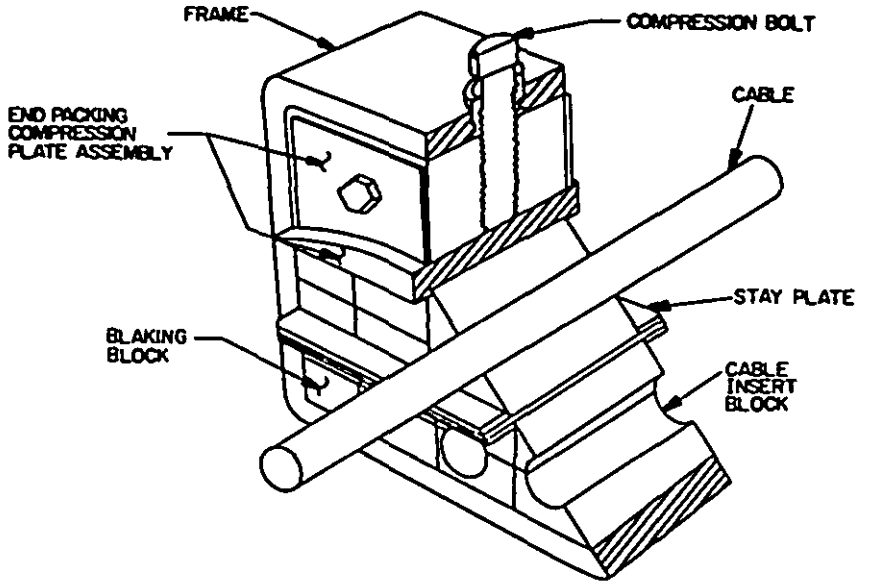
MIL-P-24705/1(SH)
7 June 1990

MILITARY SPECIFICATION SHEET

PENETRATORS, MULTIPLE CABLE, FOR ELECTRIC CABLES, RECTANGULAR METAL FRAME, FOR SQUARE-FACED BLOCKS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-P-24705.



SH 132316778

FIGURE 1. *Rectangular multiple cable penetrator - assembled view - typical.*

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, decimal tolerance is plus or minus 0.02 (0.5 mm) for dimensions with 2 decimal places or plus or minus 0.016 (0.4 mm) for dimensions with 3 decimal places.
4. Unless otherwise specified, for molded rubber parts, tolerance shall be in accordance with MIL-HDBK-149.

AMSC N/A FSC5975

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Assembly

Components required for each complete rectangular metal frame multiple cable penetrator assembly specified by this specification sheet are:

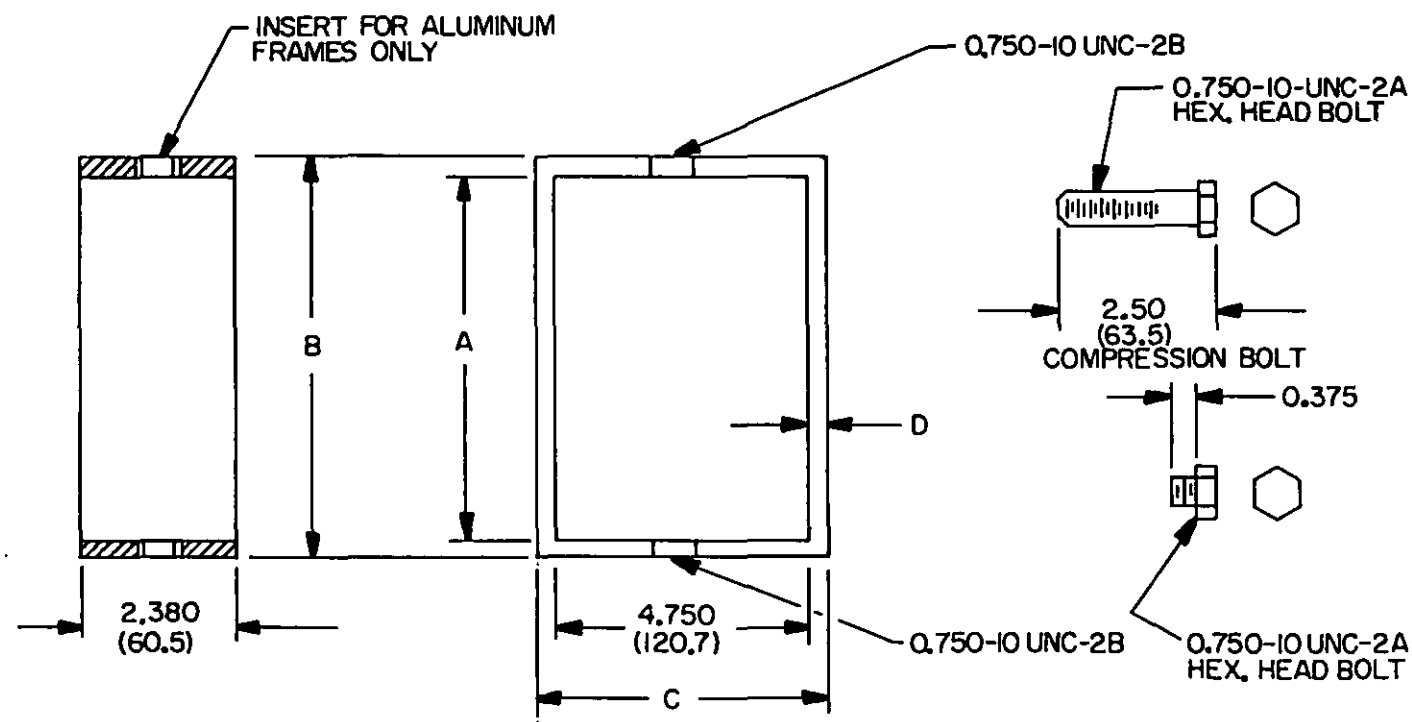
a. One rectangular metal frame with associated hardware (see figure 2)

b. The appropriate end packing compression plate assembly and all hardware (see figure 3, 4, 5, 6 and 7)

c. Stay plates (see figure 8 or 9 as required, one for each block layer)

d. Square-faced cable insert blocks, blanking insert blocks and filler inserts to fill the interior of the rectangular frame (see figures 10, 11, 12, and 13).

Construction



SH 132316779

FIGURE 2. *Rectangular frame with associated hardware.*

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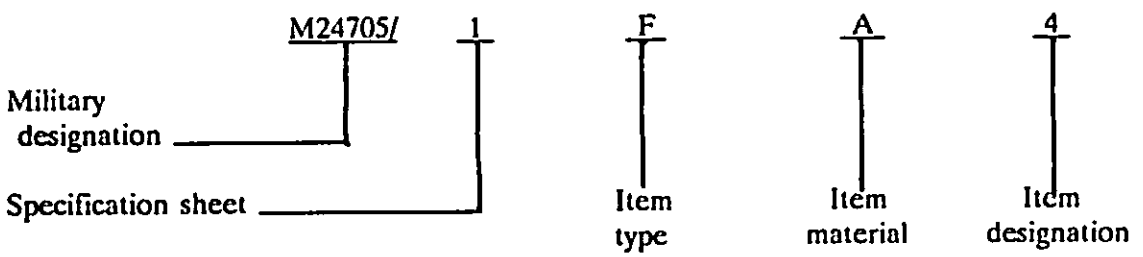
TABLE I. *Rectangular frame dimensions.*

Designator[1]	A	B	C	D
FA2	3.97 (100.8)	4.97 (126.2)	5.76 (146.3)	0.50 (12.7)
FA4	6.28 (159.5)	7.98 (184.9)		
FA6	8.59 (218.2)	9.59 (243.6)		
FA8	10.91 (277.1)	11.91 (302.5)		
F*2	3.97 (100.8)	4.72 (119.9)	5.51 (140.0)	0.38 (9.6)
F*4	6.28 (159.5)	7.03 (178.6)		
F*6	8.59 (218.2)	9.34 (237.2)		
F*8	10.91 (277.1)	11.66 (296.2)		

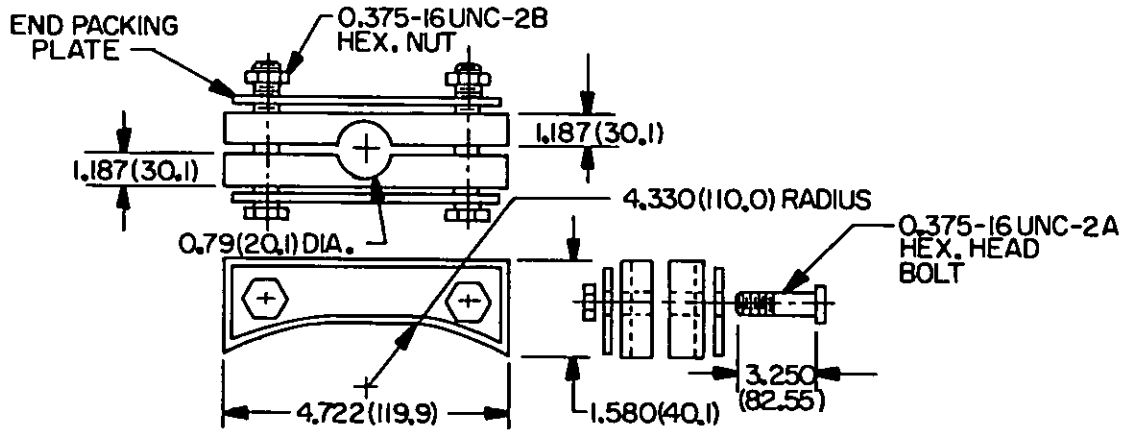
[1]The asterisk "*" represents item material. The material shall be aluminum (A), CRES (C) or steel (S) in accordance with MIL-P-24705.

PIN (see MIL-P-24705):

EXAMPLE

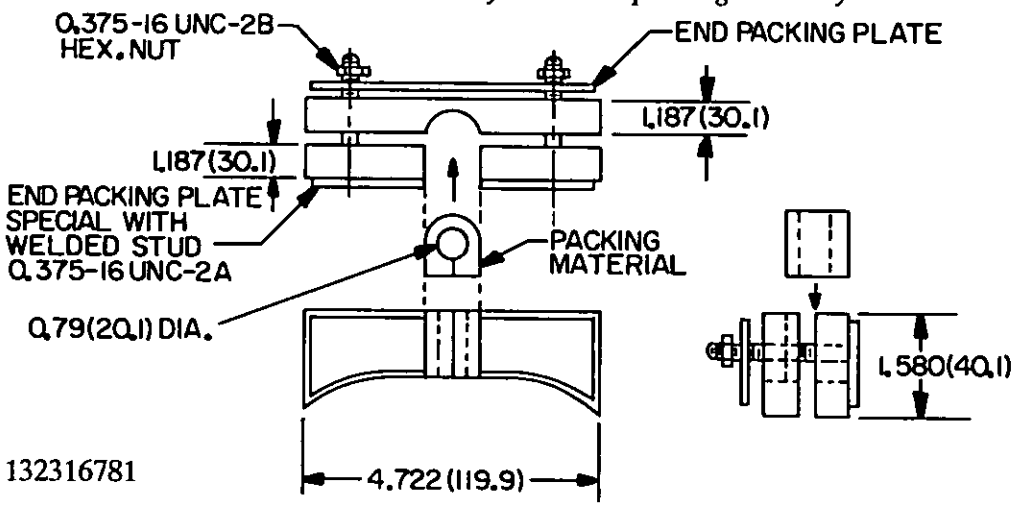


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SH 132316780

FIGURE 3. *Style 1A end packing assembly.*



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FIGURE 4. *Style 1B end packing assembly.*

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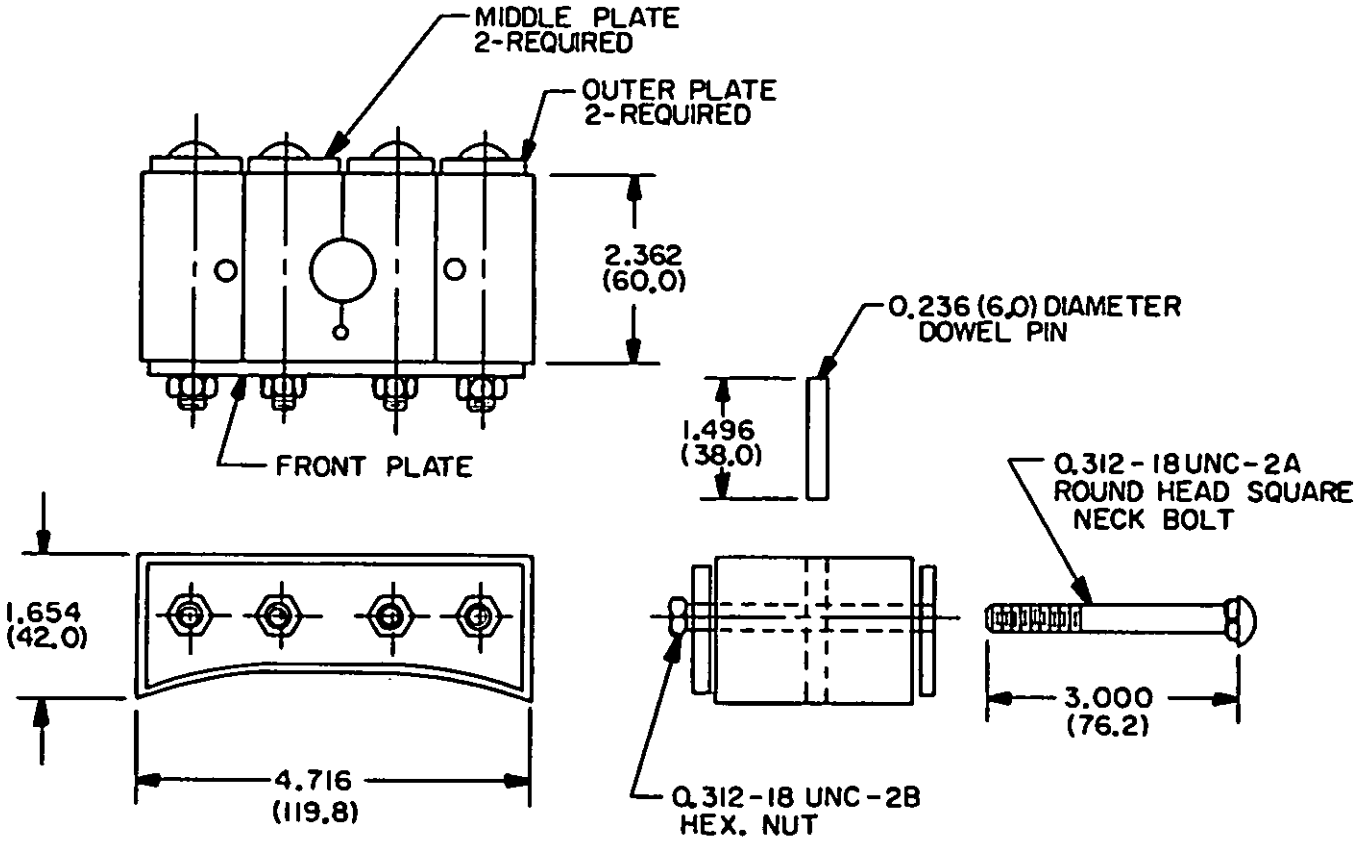
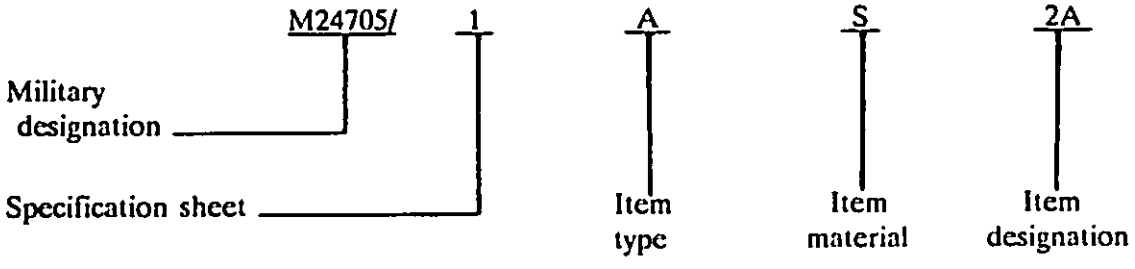


FIGURE 5. *Style 2A end packing assembly.*

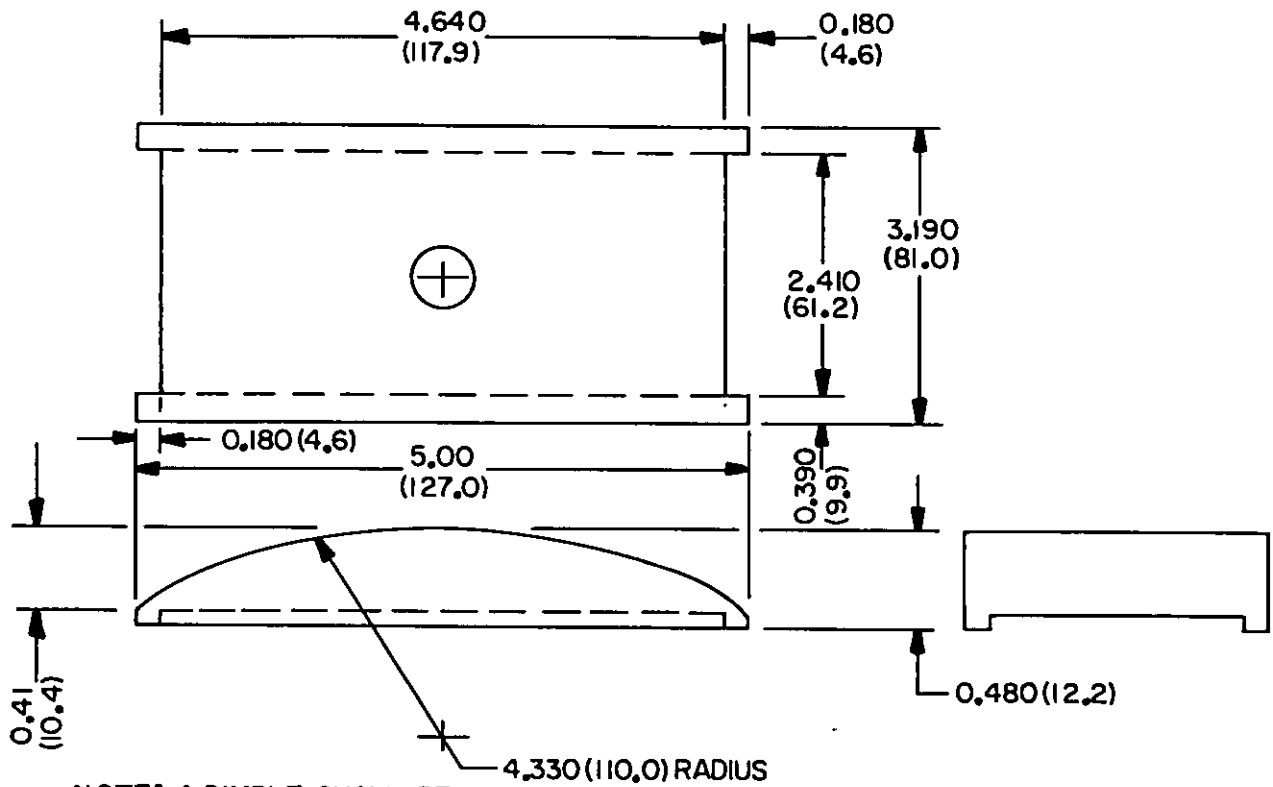
PIN (see MIL-P-24705):

The material specified for end packing assemblies in the military designation pertains to hardware only. The material shall be CRES (C) or steel (S).

EXAMPLE



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NOTE: A DIMPLE SHALL BE PROVIDED ON THE TOP CENTER OF THE COMPRESSION PLATE FACE FOR THE COMPRESSION BOLT.

SH 132316782

FIGURE 6. *Compression plate for style 1A and 1B end packing.*

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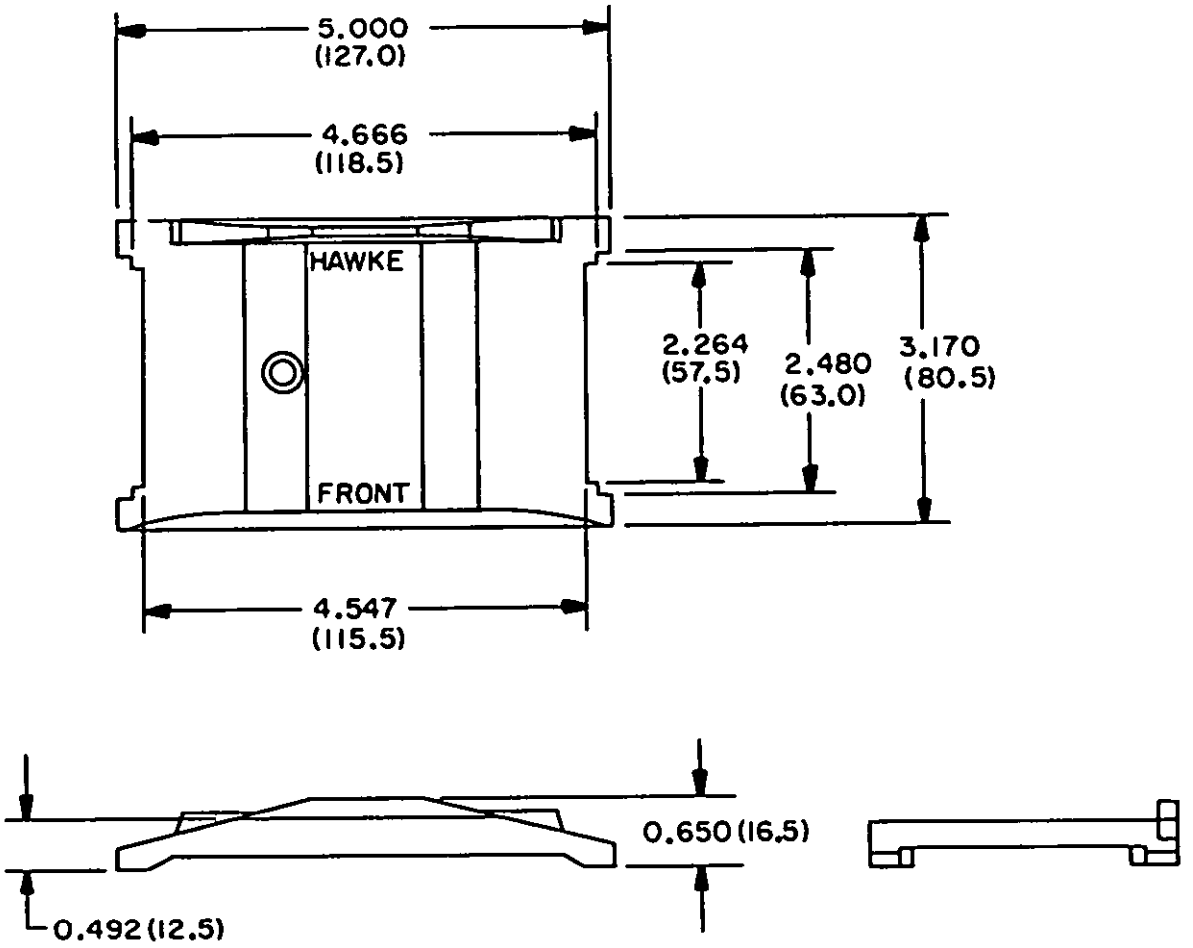
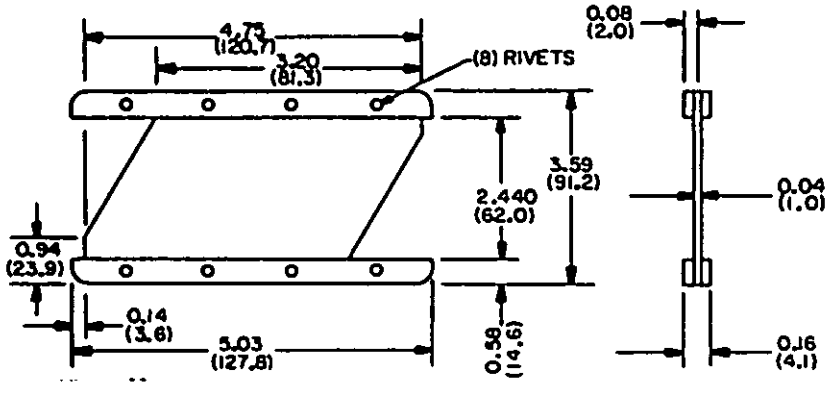


FIGURE 7. Compression plate for style 2A end packing.



SH 132316783

FIGURE 8. *Stay plate.*

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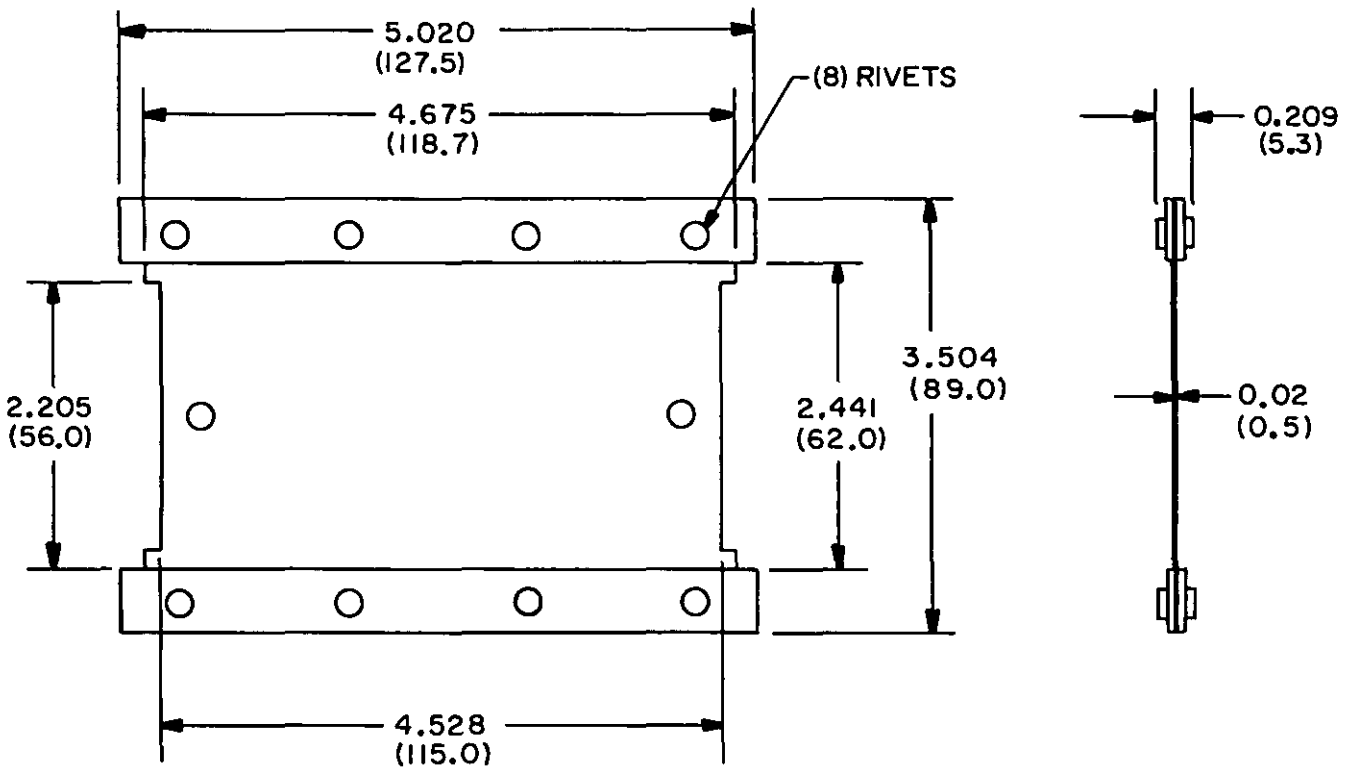


FIGURE 9. Style 2 stay plate.

PIN (see MIL-P-24705):

The material specified for stay plates in the military designation shall be aluminum (A), brass (B), CRES (C), or steel (S). The item designation shall consist of a single digit corresponding to a particular style.

EXAMPLE

M24705/ 1 S S 2

Military designation

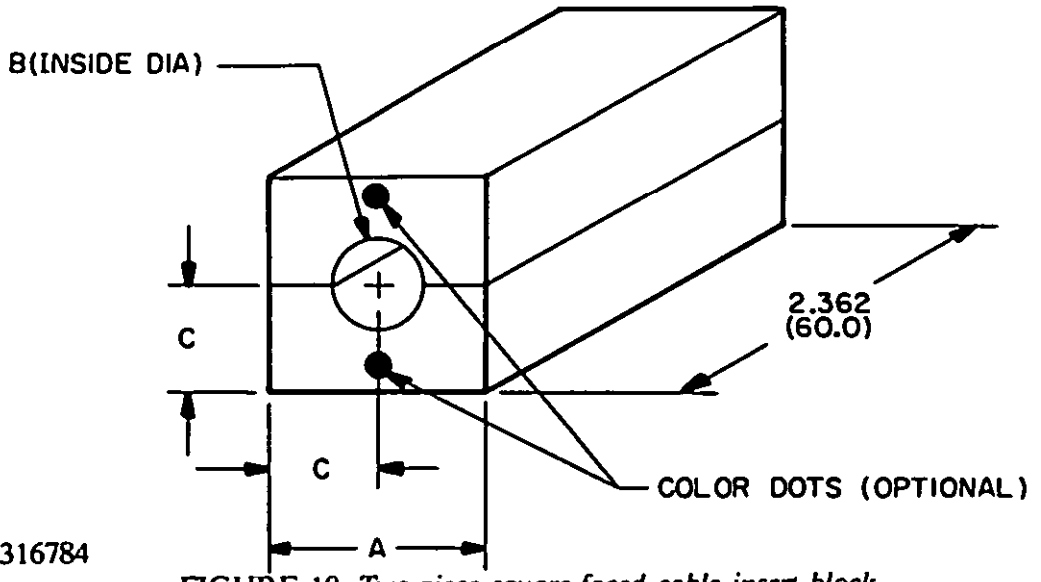
Specification sheet

Item type

Item material

Item designation

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SH 132316784

FIGURE 10. *Two-piece square-faced cable insert block.*

TABLE II. *Two-piece square-faced cable insert block dimensions.*[1]

Designator	A	B	C	Color dots
BN1503/5	0.591 (15.0)	0.158(4.02)	0.295 (7.5)	Red
BN1504		.157(4.0)		
BN1505		.197(5.0)		
BN1505/7		.237(6.03)		White
BN1506		.236(6.0)		
BN1507		.276(7.0)		
BN1507/9		.317(8.04)		Blue
BN1508		.315(8.0)		
BN1509		.354(9.0)		
BN2003/6	0.787 (20.0)	0.158(4.02)	0.394 (10.0)	Green
BN2004		.157(4.0)		
BN2005		.197(5.0)		
BN2006		.236(6.0)		
BN2006/9		.277(7.04)		Pink
BN2007		.276(7.0)		
BN2008		.315(8.0)		
BN2009		.354(9.0)		
BN2009/12		.396(10.05)		Brown
BN2010		.394(10.0)		
BN2011		.433(11.0)		
BN2012		.472(12.0)		
BN2013		.512(13.0)		
BN2013/16		.554(14.07)		Gold
BN2014		.551(14.0)		
BN2015		.591(15.0)		
BN2016		.630(16.0)		

See footnote at end of table.

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TABLE II. *Two-piece square-faced cable insert block dimensions - Continued*[1].

Designator	A	B	C	Color dots
BN3012	1.181 (30.0)	0.472(12.0)	0.591 (15.0)	
BN3012/15		.515(13.07)		Red
BN3013		.512(13.0)		
BN3014		.551(14.0)		
BN3015		.591(15.0)		
BN3015/18		.633(16.08)		White
BN3016		.630(16.0)		
BN3017		.669(17.0)		
BN3018		.709(18.0)		
BN3018/21		.752(19.10)		Blue
BN3019		.748(19.0)		
BN3020		.787(20.0)		
BN3021		.827(21.0)		
BN3021/24		.870(22.11)		Orange
BN3022		.866(22.0)		
BN3023		.906(23.0)		
BN3024		.945(24.0)		
BN4022	1.575 (40.0)	0.866(22.0)	0.787 (20.0)	
BN4022/25		.910(23.12		Red
BN4023		.906(23.0)		
BN4024		.945(24.0)		
BN4025		.984(25.0)		
BN4025/28		1.029(26.13)		White
BN4026		1.024(26.0)		
BN4027		1.063(27.0)		
BN4028		1.102(28.0)		
BN4028/31		1.148(29.15)		Blue
BN4029		1.142(29.0)		
BN4030		1.181(30.0)		
BN4031		1.220(31.0)		
BN4031/34		1.266(32.16)		Orange
BN4032		1.260(32.0)		
BN4033		1.299(33.0)		
BN4034		1.339(34.0)		
BN6031/34	2.362 (60.0)	1.266(32.16)	1.181 (30.0)	Red
BN6032		1.260(32.0)		
BN6033		1.299(33.0)		
BN6034		1.339(34.0)		
BN6034/37		1.385(35.18)		White
BN6035		1.378(35.0)		
BN6036		1.417(36.0)		
BN6037		1.457(37.0)		
BN6037/40		1.504(38.19)		Blue
BN6038		1.496(38.0)		
BN6039		1.535(39.0)		
BN6040		1.575(40.0)		

See footnote at end of table

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TABLE II. Two-piece square-faced cable insert block dimensions. - Continued 1/

Designator	A	B	C	Color dots
BN6040/43	2.362 (60.0)	1.622(41.21)	1.181 (30.0)	Orange
BN6041		1.614(41.0)		
BN6042		1.654(42.0)		
BN6043		1.693(43.0)		
BN6043/46		1.741(44.22)		Purple
BN6044		1.742(44.0)		
BN6045		1.772(45.0)		
BN6046		1.811(46.0)		
BN6046/49		1.860(47.24)		Yellow
BN6047		1.850(47.0)		
BN6048		1.890(48.0)		
BN6049		1.929(49.0)		
BN6049/52		1.978(50.25)		Green
BN6050		1.968(50.0)		
BN6051		2.008(51.0)		
BN6052		2.047(52.0)		
BN6053		2.087(53.0)		
BN6054		2.126(54.0)		
BN6054/56		2.150(54.61)		Pink
BN6055		2.165(55.0)		
BN8054	3.150 (80.0)	2.126(54.0)	1.575 (40.0)	
BN8056		2.205(56.0)		
BN8058		2.283(58.0)		
BN8060		2.362(60.0)		
BN8062		2.441(62.0)		
BN8064		2.520(64.0)		
BN8066		2.598(66.0)		
BN8068		2.677(68.0)		
BN8070		2.756(70.0)		
BN9050	3.543 (90.0)	1.969(50.0)	1.772 (45.0)	
BN9051		2.008(51.0)		
BN9052		2.047(52.0)		
BN9053		2.087(53.0)		
BN9053/56		2.137(54.27)		Red
BN9054		2.126(54.0)		
BN9055		2.165(55.0)		
BN9056		2.205(56.0)		
BN9056/59		2.256(57.29)		White
BN9057		2.244(57.0)		
BN9058		2.283(58.0)		
BN9059		2.323(59.0)		
BN9059/62		2.374(60.30)		Blue
BN9060		2.362(60.0)		
BN9061		2.402(61.0)		
BN9062		2.441(62.0)		
BN9062/65		2.493(63.32)		Orange

See footnote at end of table.

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TABLE II. Two-piece square-faced cable insert block dimensions. - Continued 1/

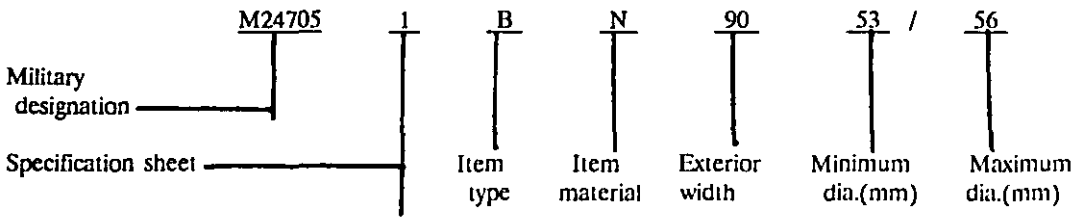
Designator	A	B	C	Color dots
BN9063	3.543 (90.0)	2.480(63.0)	1.772 (45.0)	
BN9064		2.520(64.0)		
BN9065		2.559(65.0)		
BN9065/68		2.611(66.33)		Purple
BN9066		2.598(66.0)		
BN9067		2.638(67.0)		
BN9068		2.677(68.0)		
BN9068/71		2.730(69.35)		Yellow
BN9069		2.717(69.0)		
BN9070		2.756(70.0)		

[1]Tolerance for cable insert block dimensions shall be plus 0.005 (0.13 mm), minus 0.000 (0.00 mm).

Military part number (see MIL-P-24705):

The item designation for BN15 thru BN90** square-faced cable insert blocks shall consist of four digits. The first two digits indicate the exterior width of the cable insert block in mm; the last two indicate the passage diameter in mm. The item designation for BN15 thru BN90**/** square-faced cable insert blocks with "COLOR DOTS" shall consist of six digits and a / after the fourth digit. The first two digits indicate the exterior width of the cable insert block in mm; the two digits before the / indicate the minimum diameter in mm, and the last two after the / indicate the maximum diameter in mm.

EXAMPLE



Item designation

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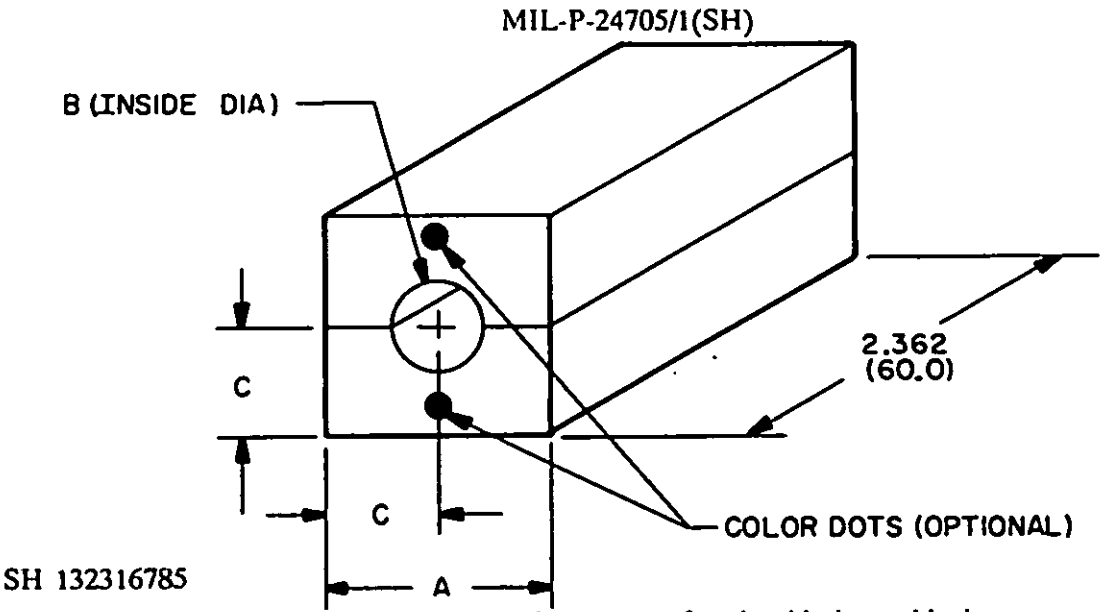


FIGURE 11. *Two-piece square-faced cable insert block.*

TABLE III. Two-piece square-faced cable insert block dimensions.[1]

Designator	A	B	C	Color dots
BN12072/75	4.724 (120.0)	2.849(72.36)	2.362 (60.0)	Red
BN12075		2.953(75.0)		
BN12075/78		2.992(76.0)		White
BN12078/81		3.110(79.0)		Blue
BN12080		3.150(80.0)		
BN12081/84		3.228(82.0)		Orange
BN12084/87		3.346(85.0)		Purple
BN12085		3.346(85.0)		
BN12087/90		3.465(88.0)		Yellow
BN12090		3.543(90.0)		
BN12090/93		3.583(91.0)		Green
BN12093/96		3.701(94.0)		Pink
BN12095		3.740(95.0)		
BN12096/100		3.819(97.0)		Brown
BN120100/103		3.976(101.0)		Gold

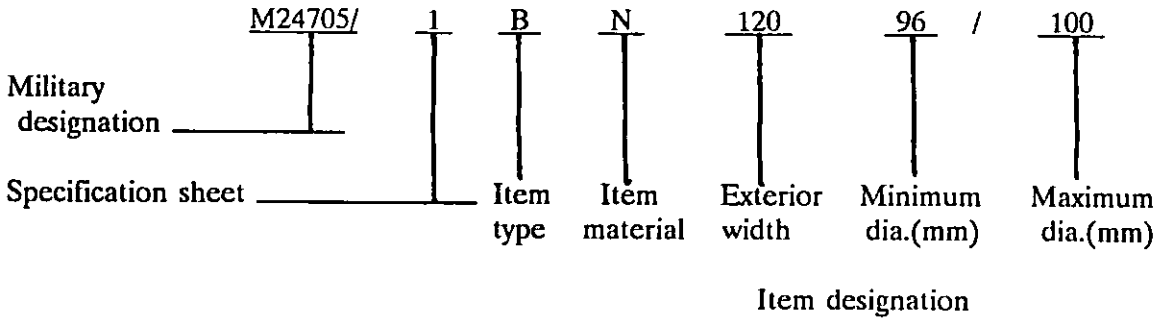
[1]Tolerance for cable insert block dimensions shall be plus 0.005 (0.13 mm), minus 0.000 (0.00 mm).

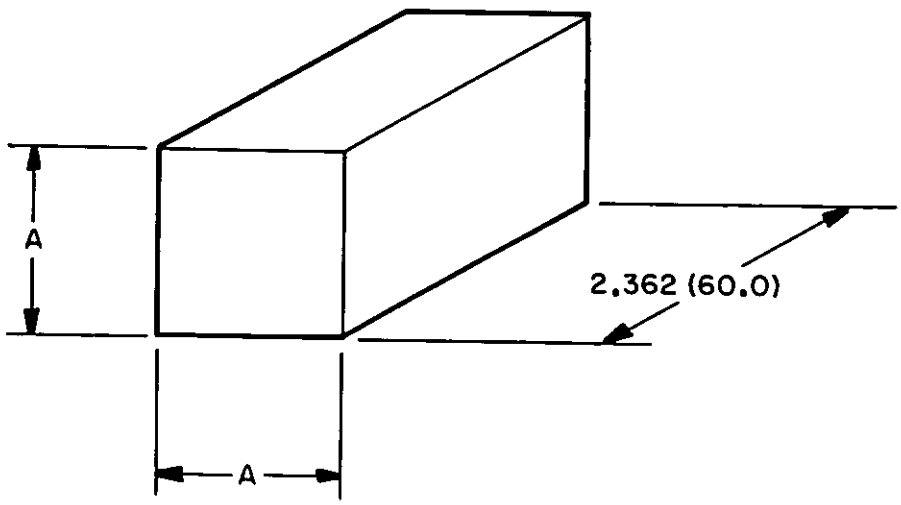
PIN (see MIL-P-24705):

The item designation for BN120** square-faced cable insert blocks shall consist of five digits. The first three digits indicate the exterior width of the cable insert block in mm; the last two indicate the passage diameter of cable insert block in mm. The item designation for BN120**/** square-faced cable insert blocks with "COLOR DOTS" shall consist of seven to nine digits and a / after the fifth or sixth digit. The first three digits indicate the exterior width of the cable insert block in mm; the two or three digits before the / indicate the minimum diameter in mm and the last two or three after the / indicate the maximum diameter in mm.

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EXAMPLE





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FIGURE 12. *Blanking insert block.*

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TABLE IV. *Blanking insert block dimensions.*[1]

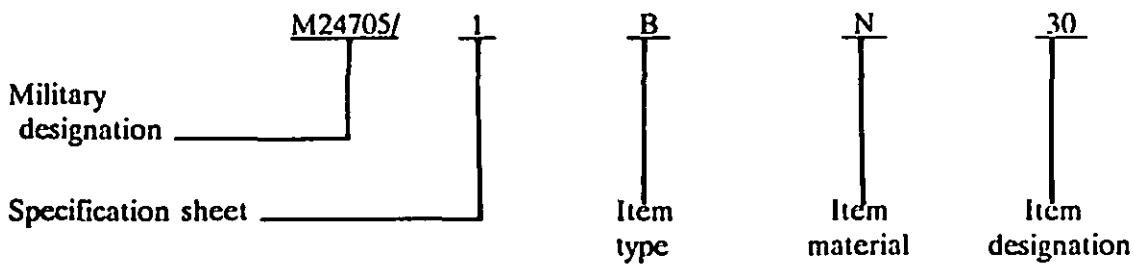
Designator	A
BN10	0.394 (10.0)
BN15	0.591 (15.0)
BN20	0.787 (20.0)
BN30	1.181 (30.0)
BN40	1.575 (40.0)
BN60	2.362 (60.0)
BN80	3.150 (80.0)
BN90	3.543 (90.0)
BN120	4.724 (120.0)

[1]Tolerance for blanking insert block dimensions shall be plus 0.005 (0.13 mm), minus 0.000 (0.00 mm).

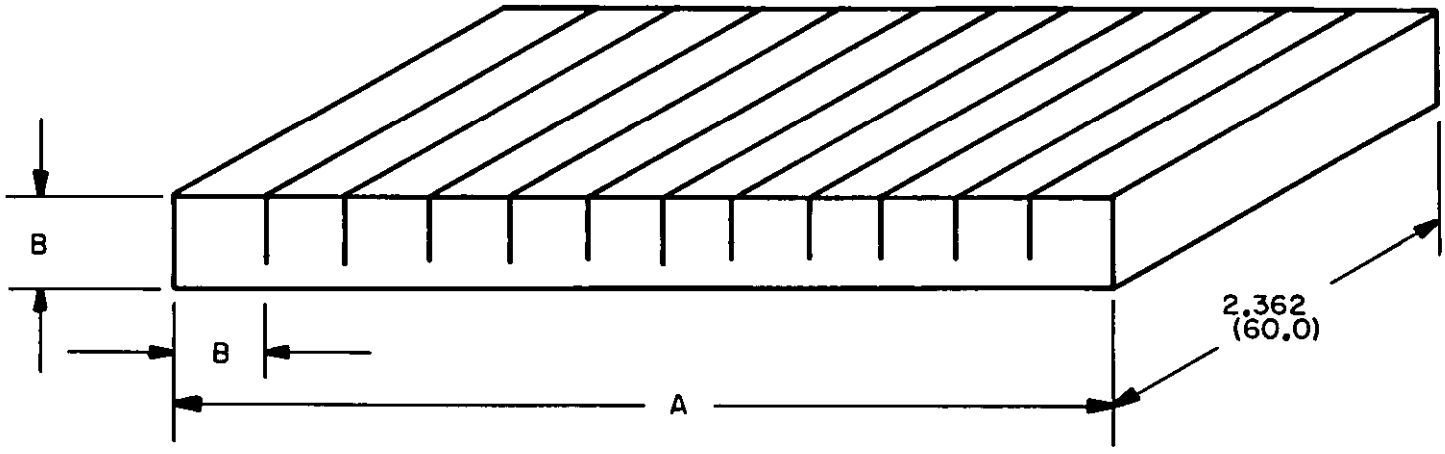
PIN (see MIL-P-24705):

The item designation for blanking insert blocks shall consist of two or three digits that indicate the exterior width of the blanking insert block in mm.

EXAMPLE



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SH 132316787

FIGURE 13. *Filler insert.*

TABLE V. *Filler insert dimensions.*[1]

Designator	A	B
IN0005	4.724 (120.0)	0.197 (5.0)
IN0010	4.724 (120.0)	0.394 (10.0)
IN1210	4.724 (120.0)	0.394 (10.0)
IN2405	4.724 (120.0)	0.197 (5.0)

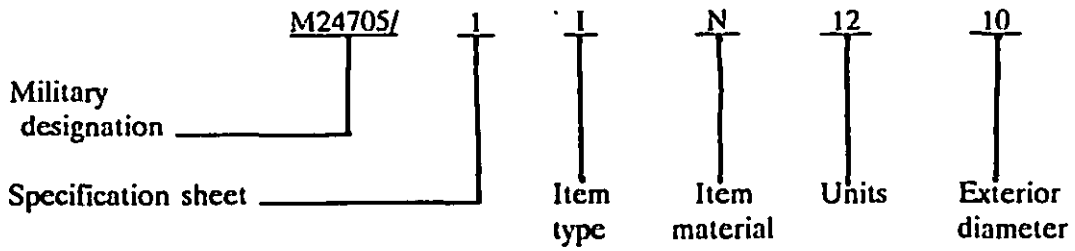
[1]Tolerance for filler insert block dimensions shall be plus 0.005 (0.13 mm), minus 0.000 (0.00 mm).

PIN (see MIL-P-24705):

The item designation for filler inserts shall consist of four digits. The first two digits indicate the quantity of filler insert units provided in the filler insert piece, the last two indicate the exterior width dimension of each filler insert unit in mm. IN0005 and IN0010 are solid filler insert pieces thus no insert units are provided.

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EXAMPLE



Item designation

REQUIREMENTS

Nonmagnetic:	When specified (see MIL-P-24705)
Durometer:	81 ± 6
Compression:	2 percent (maximum)
Expansion:	1 percent (maximum)
Sample size:	One BN60
Fluid immersion:	Plus or minus 5 (maximum)
Sample size:	Three BN40
IR signature:	Informational
Sample size:	One BN60

Identification marking: Identification marking shall be required on 20 mm exterior width cable insert blocks and larger. Passage diameter shall not be required on blanking insert blocks. On filler inserts, item designation and manufacturers name or logo shall be on the inserts, other markings are not required. Optional markings on cable insert blocks shall consist of color code dots for matching insert block halves of a selected module size, cable diameter range in inches and manufacturers catalog numbers.

Rectangular frame multiple cable penetrator test specimen (see figure 14).

The components shall be as specified below. Each component listed shall be required for each test specimen:

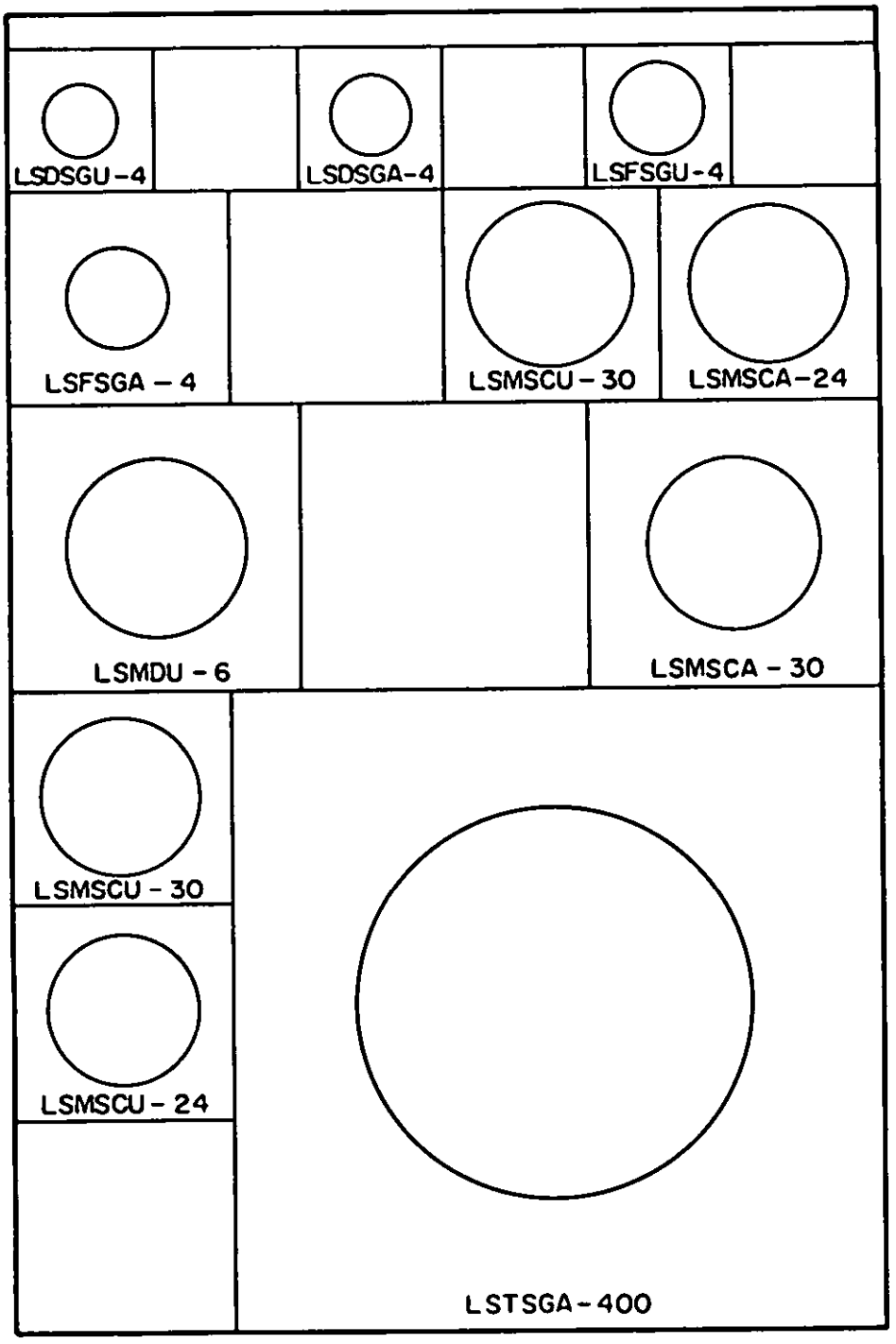
Frame:	1 - F*6
End packing assembly:	1 - E*1
Compression plate:	1 - C*
Stay plate:	3 - S*2
Two-piece square-faced cable insert blocks:	2 - BN2009/12
	1 - BN3012/16
	4 - BN4022/25

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	1 - BN9053/56
	3 - BN20
Blanking insert blocks:	2 - BN30
	1 - BN40
Filler inserts:	1 - IN0005
Cables:	LSDSGA-4, LSDSGU-4
	LSFSGA-4, LSFSGU-4
	LSMSCA-24, LSMSCU-24
	LSMSCA-30, LSMSCU-30
	LSMDU-6, LSTSGA-400
Lubricant:	MIL-S-8660

* Represents item material which may be any of the materials specified herein.

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SH 13216788

FIGURE 14. *Rectangular frame multiple cable penetrator test specimen - cable configuration.*

Preparing activity:
Navy - SH
(Project 5975-N002-1)

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